Project Workflow

E-Commerce Sales Drop Analysis

Important: these are synthetic portfolio-learning datasets, not real company data. Treat hypotheses as questions to test, not established findings.

1. Company scenario and problem statement

An e-commerce business has seen a decline in order volume and revenue in recent months. The analyst must establish when the decline occurred, quantify it, identify which products/customer groups/regions/channels contributed, and recommend evidence-based next steps.

2. Business objectives

Translate the business concern into measurable questions before opening a dashboard.

• Define the primary outcome metric and supporting metrics.

• Identify the unit of analysis and the correct grain for each table.

• Establish a defensible comparison baseline and note any limitations in the time window.

3. Data understanding

Table	Grain / role
orders_raw.csv	Order-line facts. One row represents one product line in an order; order_id is the order identifier, not necessarily unique per line in a real multi-item order. Here the synthetic dataset uses one product line per order ID.
customers_raw.csv	Customer dimension; one row per customer after deduplication.
products_raw.csv	Product dimension; one row per product after deduplication.

Before analysis, inspect row counts, distinct keys, date coverage, missingness, and relationships. Confirm that joins do not unexpectedly multiply rows.

4. Data cleaning

Use 03_Data_Cleaning_Guide.pdf and reproduce each documented step. Record the tool used, transformation, row count before/after, and validation result.

The supplied cleaned CSVs are the reference outputs. Keep IDs as text and do not silently remove source columns.

5. Metrics to define

Metric / analysis	Definition or note
Revenue and revenue % change by month	Write the exact formula, grain, date window, and denominator before reporting the result.
Order count and order-volume change	Write the exact formula, grain, date window, and denominator before reporting the result.
Average order value (revenue / distinct orders)	Write the exact formula, grain, date window, and denominator before reporting the result.
Revenue by category, product, region, segment, and channel	Write the exact formula, grain, date window, and denominator before reporting the result.
Estimated profit and profit margin (revenue less estimated cost)	Write the exact formula, grain, date window, and denominator before reporting the result.
Discount rate and quantity mix	Write the exact formula, grain, date window, and denominator before reporting the result.

Where relevant, report absolute values and percentage changes together. Make the denominator explicit for every rate.

6. Hypotheses to test

These are starting hypotheses, not findings. For each one, record evidence that would support it, evidence that would weaken it, and any extra data needed.

• The decline may be concentrated in a subset of product categories or products.

• Order volume may have fallen more than average order value, or vice versa.

• A region, customer segment, or sales channel may account for a disproportionate share of the decline.

• Discounting or product mix may have affected estimated profit even if revenue moved differently.

7. SQL investigation plan

• Start with data profiling: COUNT(*), COUNT(DISTINCT key), MIN/MAX dates, null/blank counts, and duplicate checks.

• Create a baseline query for the primary metric by period or segment.

• Use conditional aggregation, joins, CTEs, and window functions only where they answer a specific question.

• Validate every important total against a second calculation or a pivot/summary.

• Save each query with a business-question comment and record the result it produces.

8. Excel validation plan

• Use PivotTables or equivalent summaries to cross-check SQL totals.

• Check percentage change formulas and denominators.

• Inspect outliers and missing data; do not assume every unusual value is an error.

• Document any discrepancy between Excel and SQL before proceeding.

9. Power BI dashboard plan

Build the dashboard only after the data model and findings are understood. Proposed visual groups:

• KPI cards: revenue, orders, AOV, estimated profit.

• Monthly trend with comparison period.

• Category/product contribution and region/channel breakdowns.

• Slicers for date, category, region, and channel.

10. Findings and evidence log — complete after analysis

Finding #	Observation and metric	Comparison / denominator	Interpretation	Confidence / limitation
1	[Fill in after querying]	[Period / segment / baseline]	[What the evidence suggests]	[What remains uncertain]
2	[Fill in after querying]	[Period / segment / baseline]	[What the evidence suggests]	[What remains uncertain]
3	[Fill in after querying]	[Period / segment / baseline]	[What the evidence suggests]	[What remains uncertain]
4	[Fill in after querying]	[Period / segment / baseline]	[What the evidence suggests]	[What remains uncertain]

Do not copy a hypothesis into this section as a finding. A finding must be supported by an actual query, a metric, and a clearly stated comparison.

11. Recommendations and measurement

Recommendation	Finding supporting it	Owner / next step	Success KPI	Risk / guardrail
[To be completed]	[Finding #]	[Action to test]	[Metric and target to define]	[Possible downside]
[To be completed]	[Finding #]	[Action to test]	[Metric and target to define]	[Possible downside]
[To be completed]	[Finding #]	[Action to test]	[Metric and target to define]	[Possible downside]

Prioritize recommendations by expected impact, confidence, effort, and reversibility. Where causality is uncertain, propose a controlled test or further investigation.

12. Limitations and further data

Document coverage gaps, missing variables, measurement issues, small segment sizes, incomplete observation windows, and synthetic-data limitations.

Do not claim a causal root cause unless the design supports it. Observational data usually supports associations and prioritized hypotheses, not proof of causation.

13. Interview simulation

• Why did you choose the comparison period?

• How did you validate revenue and order counts?

• Why should average order value be considered with order volume?

• How did you identify which segments contributed to the total change?

• What data would you request before claiming a root cause?

Prepare a 60–90 second summary using this structure: business problem → data and method → key evidence → recommendation → limitation. Complete it after you have your own findings.

Practice follow-ups by explaining why you chose each metric, how you validated it, what alternative explanation exists, and what you would investigate next.

14. Final project checklist

• I can explain the business problem and why each metric matters.

• I can reproduce the cleaned dataset from the raw data.

• I can explain table grain, keys, joins, and metric denominators.

• I have saved readable SQL queries and cross-checked important totals.

• My dashboard answers the business questions rather than merely displaying charts.

• Every finding has evidence, a comparison, and a stated limitation.

• My recommendations connect to findings and define how success will be measured.